Poissonian Blurred Hyperspectral Imagery Denoising Based on Variable Splitting and Penalty Technique

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***Abstract*— Poisson noise is one of the significant sources of noise present in hyperspectral imagery (HSI). In most of the existing denoising methods, Poisson noise is first transformed into Gaussian noise through the Anscombe transform and then removed. However, the use of Anscombe transform can give rise to transform errors that affect the final denoising results. In addition, blurs often contaminate the HSI during the imaging procedure, which makes it more difficult to remove the Poisson noise. In view of the above problems, under the maximum a posteriori (MAP) model, we propose a Poissonian blurred HSI denoising based on variable splitting and penalty technique (named as VSPT) to directly remove the Poissonian blurred HSI noise without using the Anscombe transform. By finding the minimum value of the negative logarithmic Poisson log-likelihood combined with the total variation (TV), the proposed method transforms the problem into two subproblems, which are easier to solve: 1) a TV regularized deconvolution problem and 2) an ordinary convex optimization problem. The experimental results show that the proposed VSPT method can effectively remove Poisson noise in HSI contaminated by blurs during the imaging procedure.**

***Index Terms*— Denoising, hyperspectral imagery (HSI), maximum a posteriori (MAP) estimation, Poisson noise, variable splitting and penalty technique (VSPT).**

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I. Introduction

HYPERSPECTRAL data are made up of dozens or even hundreds of images of one observation scene in different spectral bands. So, they can provide a large amount of extra information, which the human eye cannot capture from real scenes [1]. Therefore, hyperspectral imagery (HSI) analysis has attracted a great deal of interest from researchers in various broad application domains, such as feature extraction [2], classification [3], unmixing [4], pansharpening [5], subpixel mapping [6], target detection [7], compression [8], and anomaly detection [9].

However, there are limitations on hardware devices. As a result, the acquired HSI usually suffers from varied noise including Gaussian noise, Poisson noise, and stripes noise [10]. Furthermore, in many cases, the blurs will also be accompanied by noise in the HSI to form a mixture of different kinds of degradations [11]. These degradations affect the visual appearance and the subsequent application of HSI. In view of this, there is a high necessary for the removal of noise and improvement of the quality before using HSI in different applications, since it is often economically impractical to rely only on hardware to improve the HSI quality. Some methods based on image processing are commonly used to enhance the image quality.

As a preprocessing operation in practical HSI exploitation, denoising can improve the subsequent HSI analysis. However, the HSI denoising task is challenging owing to the presence

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of different noise sources. HSI has the following common types of noises: signal independent noise, signal-dependent noise, sparse noise, pattern noise, and a mixture of them [12]. Among them, the signal independent noise adopts the Gaussian noise to model the thermal and quantization noise in HSI. As the Gaussian assumption can considerably simplify the analysis, it becomes the most common noise assumption in most denoising methods [13].

Many denoising methods have been developed including the methods of transform, the methods of low rank, the methods of total variation (TV) regularization, the methods of tensor, and the methods of deep learning. The methods that are based on transform mainly adopt principal component analysis (PCA) and maximum noise fraction (MNF) transform [14]. The work of [15] was the first proposed method, which used the low-rank structure [16], [17], [18] of HSI to remove the noise. It used the low-rank property along the spectral mode to arrange a 3-D HSI into a 2-D matrix lexicographically. He et al. [19] set up the patchwise low-rank matrix approximation to consider the different intensities of Gaussian noise in different bands.

The work of Zhuang and Bioucas-Dias [20] proposed fast algorithms for denoising and inpainting of HSI based on their low-rank and self-similarity characteristics. Recently, the TV regularization has also been widely adopted in the HSI restoration [21], [22], [23]. The nuclear norm, TV regularization, and L1-norm were integrated into a unified framework in [21]. Among them, the TV regularization was used to explore the HSI's spatial piecewise smooth structure. As the low rank approximation and sparsity can be easily incorporated into the framework of tensor, the researchers directly applied the methods of tensor to degraded HSI [24], [25], [26]. For example, tensor Tucker decomposition was used to describe the global correlation among all bands [24]. As mentioned above, low-rank regularization has been used to explore the spectral correlation, but these methods are often quite complex and difficult to deal with some practical problems in time [27], [28].

Recently, the methods of deep learning have achieved success in diverse tasks of image processing. Many researchers have applied the deep learning methods to HSI denoising [29]. For instance, convolutional neural network (CNN) has been introduced into HSI recovery [30]. However, the main challenge of the CNN model to HSI recovery is to explore spatial–spectral information [31], [32]. In [33], the spectrum-aware and transferable architecture search (STAS) is utilized to automatically choose the convolutions of the adaptive spatial–spectral information exploration of HSI. At the same time, it takes a significant amount of manpower and time to obtain enough data needed to train the deep neural network [34]. Therefore, another current research trend is to explore the recovery of the degraded HSI in the absence of training data [30], [35]. In addition, there may be impulse noises, such as salt and pepper noise, missing pixels, missing lines, and other outliers in the HSI acquisition process. Most of these noises are sparse, and therefore, they are usually classified as sparse noises.

In order to remove this sparse noise, researchers often use sparsity techniques or sparse and low-rank decomposition techniques. For example, artifacts may arise during the HSI imaging procedure, which are considered as pattern noise [36]. Stripe noise, which is usually caused by calibration errors and detector sensitivity variations, is a type of pattern noise that may occur when the system scans the target line by line [37], [38]. The above three types of noise can be directly modeled in the commonly used additive models of HSI. The mixture of those noise can be removed by representing the unprocessed collected data as the sum of clean data and additive noise [39], [40].

The development of sensor techniques will increase the number of spectral bands and decrease the spectral bandwidth. This means that when everything else remains the same, each spectral channel will receive fewer photons, resulting in a higher level of signal independent noise modeled by the Poisson distribution (Poisson noise) [20]. Therefore, Poisson noise is becoming a more and more important concern in real HSI [41], [42], [43]. However, recently, the analysis for hyperspectral Poisson noise got stuck, and few works are designed for this kind of noise [41], [42], [43], and the common denoising methods based on the additive model cannot deal with Poisson noise without converting it into Gaussian noise. This is because the Poisson noise does not depend on data and is neither additive nor multiplicative [44].

So to remove the Poisson noise, most existing methods will first adopt the Anscombe transform (a variance-stabilizing transformation) [45], [46] to convert Poisson noise into approximate Gaussian noise. The traditional additive Gaussian noise denoising methods are then used to remove the noise. Finally, to obtain the estimated value of the HSI of interest, the denoised HSI will adopt inverse transform finally [47], [48]. The challenge of these forward and inverse transforms inevitably involves many unstable factors.

Recently, variance-stabilizing transformation has been investigated as a denoising method for Poisson noise [49], [50], especially, when the count regime is relatively low, that is, for low intensity signals generated under low signal-to-noise ratio (SNR) [13]. In addition, factors such as atmospheric turbulence or imaging equipment motion are unavoidable during the imaging procedure, which result in blurred contamination during the HSI imaging process, it makes removing the noise difficult.

To solve the issues, we propose a method capable of directly removing noise from Poissonian blurred HSI. We call it as Poissonian-blurred HSI denoising based on variable splitting and penalty technique (VSPT). In this method, we model the latent image given the blur by maximum a posteriori (MAP) estimation in the Bayesian probabilistic framework. Subsequently, we look for the minimum value of the combination of the negative logarithmic Poisson log-likelihood and the TV [51]. The variable splitting and penalty method is then used to transform the above model into two simpler subproblems to produce the final denoising result: 1) solving a TV regularized deconvolution problem and 2) convex optimization

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problem. The main contributions of the proposed VSPT can be summarized as follows.

1) The MAP estimation in the Bayesian probability framework is used to model image with Poisson noise. This avoids the errors induced by the Anscombe transform in converting Poisson noise to approximate additive Gaussian noise. It consequently improves the success rate of removing the Poisson noise.
2) The Poisson noise can be removed when blurs exist in the imaging process. In other words, the degradations caused by the Poisson noise and blur could be removed.
3) Experimental results demonstrate that compared with the current HSI denoising methods, the proposed VSPT could achieve better denoising results.

The remainder of this article is organized as follows. Section II summarizes the proposed methodology. The experimental results and discussions are shown in Section III. Finally, Section IV concludes our work.

II. Methodology

A. Hyperspectral Poisson Noise and Blur Degradation Model

A hyperspectral Poisson noise and blur degradation model is proposed in this section. It is supposed that the HSI is contaminated by the blurs during the imaging procedure, which is mathematically reflected in the convolution of the original HSI and the point spread function (PSF) [51]. At the same time, the blurred HSI is corrupted by the Poisson noise. Therefore, we want to recover a Poissonian blurred HSI. A model is proposed to describe the above problems as follows:

$$\mathbf{y} = \text{Poisson}(\mathbf{bx}) \tag{1}$$

where $\mathbf{x}$ and $\mathbf{y}$ denote one of the bands of the clean HSI and one of the bands of the Poissonian blurred HSI, respectively. Their size is both $M \times N$, where M and N are the spatial dimensions of the image. A blurring (or convolution) matrix form of the PSF is denoted by $\mathbf{b}$, which can be measured in advance. The Poisson$(\cdot)$ function is the Poisson noise degradation process, which is neither additive nor multiplicative [44].

In order to avoid using the Anscombe forward transform and inverse transform in the case of Poisson noise and blur, the MAP model is used here to solve the noise removal problem of Poissonian blurred HSI, as shown in the following equation:

$$\mathbf{x} = \arg\max_{\mathbf{x}} P(\mathbf{x}|\mathbf{y}) = \arg\max_{\mathbf{x}} P(\mathbf{y}|\mathbf{x})P(\mathbf{x}) \tag{2}$$

where $P(\mathbf{y}|\mathbf{x})$ is the probabilistic distribution of Poisson noise and $P(\mathbf{x})$ is the probabilistic distribution of the clean HSI. Equation (2) is solved by minimizing $-\log P(\mathbf{y}|\mathbf{x})P(\mathbf{x})$, which can be converted into the following problem:

$$\mathbf{x} = \arg\min_{\mathbf{x}} \left\{ -\ln\left[P(\mathbf{y}|\mathbf{x})\right] - \ln[P(\mathbf{x})] \right\}. \tag{3}$$

The noise term is considered to follow the Poisson distribution. Therefore, it can be formalized as

$$P(\mathbf{y}|\mathbf{x}) = \prod_{i=1}^{MN} \frac{\exp\left[-(\mathbf{bx})_i\right](\mathbf{bx})_i^{\mathbf{y}_i}}{\mathbf{y}_i!}. \tag{4}$$

Equation (5) can be derived as follows by finding the negative logarithm of (4):

$$-\ln\left[P(\mathbf{y}|\mathbf{x})\right] = \sum_{i=1}^{MN} \left\{ (\mathbf{bx})_i - \mathbf{y}_i \ln\left[(\mathbf{bx})_i\right] \right\} \tag{5}$$

where i denotes the pixel index and MN represents that the total number of pixels is $M \times N$ in each variable.

The PSF is in low frequency, while the noise is mainly concentrated in high frequency. Thus, it could be very easily amplified during the process of deblurring (deconvolution). We adopt the TV regularization to model the term $-\ln[P(\mathbf{x})]$ to get a more stable solution, i.e.,

$$-\ln[P(\mathbf{x})] = \sum_{i=1}^{MN} \sqrt{(\nabla_1 \mathbf{x})_i + (\nabla_2 \mathbf{x})_i} \tag{6}$$

where ∇_1 and ∇_2 are the convolution matrices of the derivative operators in the horizontal and vertical directions, respectively.

Subsequently, to place the nonnegative constraint on the estimate, we define a function $\Phi(\mathbf{x})$ as

$$\Phi(\mathbf{x}) = \begin{cases} 0, & \text{if } \mathbf{x} \in \mathrm{R}_{\mathbf{N}}^{+} \\ +\infty, & \text{if } \mathbf{x} \notin \mathrm{R}_{\mathbf{N}}^{+}. \end{cases} \tag{7}$$

The final hyperspectral Poisson noise and blur degradation model can be written as

$$\mathbf{x} = \arg\min_{\mathbf{x}} \frac{\lambda}{2} \sum_{i=1}^{MN} \left\{ (\mathbf{bx})_i - \mathbf{y}_i \ln\left[(\mathbf{bx})_i\right] \right\} + \sum_{i=1}^{MN} \sqrt{(\nabla_1 \mathbf{x})_i + (\nabla_2 \mathbf{x})_i} + \Phi(\mathbf{x}) \tag{8}$$

where λ is a regularization parameter, which is determined by the noise and blur levels of the degraded image.

B. Variable Splitting and Penalty Technique

Due to its good robustness, variable splitting and penalty method [52] is used to optimize the hyperspectral Poisson noise and blur degradation model in (8). By introducing an auxiliary variable $\mathbf{w}_0$ and a penalty parameter β_1, (8) can be expressed as

$$(\mathbf{x}, \mathbf{w}_0) = \arg\min_{(\mathbf{x},\mathbf{w}_0)} \frac{\lambda}{2} \sum_{i=1}^{MN} \left\{ (\mathbf{w}_0)_i - \mathbf{y}_i \ln\left[(\mathbf{w}_0)_i\right] \right\} + \frac{\beta_1}{2} ||\mathbf{bx} - \mathbf{w}_0|| + \sum_{i=1}^{MN} \sqrt{(\nabla_1 \mathbf{x})_i + (\nabla_2 \mathbf{x})_i} + \Phi(\mathbf{x}) \tag{9}$$

where $\beta_1 = \beta_1 * k$ is increased by multiplying a factor $k > 1$. The solution of (9) converges to that of (8) as β_1 approaches positive infinity.

For each value of β_1, (9) is split into the following two subproblems for $\mathbf{x}$ and $\mathbf{w}_0$, which can be solved iteratively.

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Algorithm 1 Solving for the $\mathbf{x}$ Subproblem

1. **Input:** $\mathbf{y}$, $\mathbf{b}$.
2. **Initialize:** Set $\mathbf{w}_j^{(0)} = \mathbf{y}; \beta_2 = \beta_{2_initial}$
3. **While** not converged **do**
4. $l = l + 1$
5. Compute $\mathbf{x}^{(l+1)}$ by solving (13) using $\mathbf{w}_j^{(l+1)} = \mathbf{w}_j^{(l)}$
6. Compute $\mathbf{w}_j^{(l+1)}$ by solving (12) using $\mathbf{x}^{(l+1)} = \mathbf{x}^{(l)}$
7. Check the convergence conditions: $\beta_2 = \beta_{2_max}$
8. **End while**
9. **Output:**HSI $\mathbf{x}^{(l+1)}$

1) Solving for the $\mathbf{x}$ *Subproblem:* Fixing $\mathbf{w}_0$, minimizing (9) with respect to $\mathbf{x}$ equates to solve the following problem:

$$\mathbf{x}^{(l+1)} = \arg\min_{\mathbf{x}} \frac{\beta_1}{2} \left\| \mathbf{bx} - \mathbf{w}_0^{(l)} \right\|_2^2 + \sum_{i=1}^{MN} \sqrt{(\nabla_1 \mathbf{x})_i + (\nabla_2 \mathbf{x})_i} + \Phi(\mathbf{x}) \tag{10}$$

where l denotes the iteration index.

In frequency domain, (10) can be solved effectively as to each given β_1

$$F^{(l+1)}(\mathbf{x}) = \frac{\boldsymbol{\beta}_1 F^*(\mathbf{b}) F(\mathbf{w}_0) + \boldsymbol{\beta}_2 \sum_j F^*(\nabla_j) F^{(l)}(\mathbf{w}_j)}{\boldsymbol{\beta}_1 F^*(\mathbf{b}) F(\mathbf{b}) + \boldsymbol{\beta}_2 \sum_j F^*(\nabla_j) F(\nabla_j)} \tag{11}$$

where $\beta_2 = \beta_2 * k$ is a penalty parameter for solving the $\mathbf{x}$ subproblem, $F(\cdot)$ and $F^*(\cdot)$ are the discrete Fourier transform operator and its conjugate, respectively, ∇_j $(j = 1, 2)$ represent ∇_1 and ∇_2, respectively, and $\mathbf{w}_j$ can be determined by the following shrinkage equation:

$$\left[\mathbf{w}_j^{(l+1)}\right]_{\mathbf{i}} = \max\left(M_i - \frac{1}{\beta_2}, 0\right) \frac{\left[\nabla_j \mathbf{x}^{(l+1)}\right]_i}{M_i} \tag{12}$$

where $M_i = (\sum_{j=1}^{2} [\nabla_j \mathbf{x}^{(l+1)}]_i^2)^{1/2}$.

Subsequently, we can obtain $\mathbf{x}$ by inverse Fourier transform $F^{-1}(\cdot)$ as

$$\mathbf{x}^{(l+1)} = \max\left\{ F^{-1}\left[F^{(l+1)}(\mathbf{x})\right], \mathbf{0} \right\} \tag{13}$$

where $\mathbf{0}$ is an MN-dimensional zero vector.

The process of solving the $\mathbf{x}$ subproblem is presented in Algorithm 1.

2) Solving for the $\mathbf{w}_0$ *Subproblem:* Fixing $\mathbf{x}$ and extracting all terms from (9) containing $\mathbf{w}_0$, we can get

$$\mathbf{w}_0^{(l+1)} = \arg\min_{\mathbf{w}_0} \frac{\lambda}{2} \sum_{i=1}^{MN} \left\{ (\mathbf{w}_0)_i - \mathbf{y}_i \ln\left[(\mathbf{w}_0)_i\right] \right\} + \frac{\beta_1}{2} \left\| \mathbf{w}_0 - \mathbf{bx}^{(l+1)} \right\|_2^2. \tag{14}$$

Suppose that there is a loss function $\sigma(a)$, i.e.,

$$\sigma(a) = \lambda a - \lambda y_i \ln(a) + \boldsymbol{\beta}_1 \left[a - (\mathbf{bx})_i\right]^2 \tag{15}$$

and we find its first and second derivatives as follows:

$$\frac{d[\sigma(a)]}{da} = \lambda - \frac{\lambda y_i}{a} + 2\beta_1 a - 2\beta_1 (\mathbf{bx})_i \tag{16}$$

$$\frac{d^2[\sigma(a)]}{da^2} = \frac{\lambda y_i}{a^2} + 2\beta_1. \tag{17}$$

Algorithm 2 Proposed VSPT

1. **Input:** Poissonian blurred HSI $\mathbf{y}$, PSF $\mathbf{b}$, regularization parameter λ.
2. **Initialize:** Set parameters k, $\beta_{1_\max}$ and $\beta_{2_\max}$; $\mathbf{w}_0^{(0)} = \mathbf{y}$; $\mathbf{w}_j^{(0)} = \mathbf{y}$; $\beta_1 = \beta_{1_\text{initial}}$; $\beta_2 = \beta_{2_\text{initial}}$
3. **While** not converged **do**
4. $l = l + 1$
5. Compute $\mathbf{x}^{(l+1)}$ using Algorithm 2 with $\mathbf{w}_0^{(l+1)} = \mathbf{w}_0^{(l)}$
6. Compute $\mathbf{w}_0^{(l+1)}$ using (19) with $\mathbf{x}^{(l+1)} = \mathbf{x}^{(l)}$
7. Check the convergence conditions: $\beta_1 = \beta_{1_\max}$
8. **End while**
9. **Output:**Final HSI $\mathbf{x} = \mathbf{x}^{(l+1)}$

Setting the first-order derivative given by (16) to zero, we have

$$2\beta_1 a^2 + [\lambda - 2\beta_1 (\mathbf{bx})_i] a - \lambda \mathbf{y}_i = 0. \tag{18}$$

Since $a \geq 0$, $-\lambda \mathbf{y}_i \leq 0$, and $d^2[\sigma(a)]/da^2 > 0$, we can get the only feasible solution of $a = \arg\min_a \sigma(a)$, which is the nonnegative root of (18).

The solution can be obtained by using the root formula as follows:

$$\mathbf{w}_0^{(l+1)} = \frac{\beta_1}{4} \left\{ 2\beta_1 \mathbf{bx}^{(l+1)} - \lambda + \sqrt{\left(\lambda - 2\beta_1 \mathbf{bx}^{(l+1)}\right)^2 + 8\lambda\beta_1 \mathbf{y}} \right\}. \tag{19}$$

In the case of fixed $\mathbf{x}$, the solution is achieved by solving MN independent problems $(\mathbf{w}_0)_i = \arg\min_{(\mathbf{w}_0)} \sigma[(\mathbf{w}_0)_i]$ simultaneously.

The process of the proposed VSPT is presented in Algorithm 2.

III. Experimental Results and Discussion

Simulated and real data experiments are presented in this section to demonstrate the performance of the proposed VSPT. The experimental results are evaluated both quantitatively and visually. There are six HSI denoising methods as benchmarks for comparison, i.e., the block-matching and 4-D filtering (BM4D) [53], the low-rank matrix recovery (LRMR) [15], the noise-adjusted iterative low-rank matrix approximation (NAILRMA) [19], the TV-regularized low-rank tensor decomposition (LRTDTV) [24], the fast hyperspectral denoising (FastHyDe) [20], and the factor group sparsity-regularized nonconvex low-rank approximation (FGSLR) [54]. All experiments are carried out using MATLAB R2018a on a desktop with 8-GB RAM and Intel Core i5-5350U CPU at 1.80 GHz. To achieve a fair comparison, all the benchmark codes are given by their authors and the parameters are fine-tuned by default or are set to the values described in their papers. For our proposed VSPT, we set $\beta_{1_\text{initial}} = 2^8$, $\beta_{1_\max} = 2^{14}$, $\beta_{2_\text{initial}} = 2^4$, $\beta_{2_\max} = 2^{10}$, and $k = 2$ empirically, and in accordance with the noise and blur level of the degraded image, we adjust the parameter λ between 1 and 1000.

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A. Simulated Experiments

Two public representative hyperspectral datasets—the Pavia city center data and the Washington DC Mall data [54]—are employed to evaluate the denoising methods. The Pavia city center data were collected by the reflective optics system imaging spectrometer (ROSIS), and the Washington DC Mall data were collected by the hyperspectral digital imagery collection experiment (HYDICE) sensor. Subregion with sizes of 256 × 256 × 191 and 256 × 256 × 191 is extracted for Pavia city center data and Washington DC Mall data, respectively. During the real imaging process, Poisson noise is one of the significant noise sources in HSI, and HSI is also polluted by blur. In order to simulate these scenarios simultaneously, three different noise and blur cases are compared. We normalize the gray value of each band of the HSI between [0, 1] before the simulation process. After the restoration process, we stretch the gray values of each band to the original level. The process of obtaining the simulation data is as follows.

First, the convolution is applied to obtain the blurred HSI. We assume the PSF is known and choose light Gaussian blur kernel with size 8 × 8 and standard deviations equal to 3 for simulating the blur generated in the imaging process. Second, the different levels of noise are applied to the blurred HSI. Before adding the Poisson distributed noise, the blurred HSI is scaled to a higher magnitude by a factor a [55]

$$\mathbf{y} = \text{Poisson}\big[(\mathbf{bx})/a\big]a. \tag{20}$$

The higher the value of a, the higher the level of Poisson noise. We consider $a = 1.0 \times 10^{10}$ to test the performance of different denoising methods.

As aforementioned, most conventional methods use the Anscombe transform to convert Poisson distributed noise into approximately Gaussian additive noise and then remove it. Therefore, for comparison, we use the Anscombe transform $\mathbf{y}^{\sim} := 2\sqrt{\mathbf{y} + (3/8)}$ and carry out inverse transform on the denoised signal to obtain the estimated value of the signal of interest [45].

1) Visual Quality Comparison: As there are many bands in hyperspectral data, we select one band in each dataset to present the visual effects of the different denoising methods. We take the value of λ as 70 in Pavia city center data and the value of λ as 42 in Washington DC Mall data. Figs. 1 and 2 present the visual effects of the different denoising methods for the two simulated datasets. The original HSI of band 51 of the Pavia city center data and the simulated degraded image are presented in Fig. 1(a) and (b), respectively. The original image of band 11 of the Washington DC Mall data and the simulated degraded image are presented in Fig. 2(a) and (b), respectively. Figs. 1(c)–(i) and 2(c)–(i) display the results of different denoising methods. The area of obvious improvement has been enlarged.

As shown in Figs. 1(b) and 2(b), there is almost no noticeable visual information in the degraded bands due to the simulation of a higher level of Poisson noise and the blurring effects of the imaging process. As shown in Figs. 1(c) and 2(c), it is obvious that the BM4D has the worst visual performance. The denoising results of BM4D are oversmoothed and blurred, and a large amount of detailed information is lost. As shown in Figs. 1(d)–(h) and 2(d)–(h), although the conventional methods provide visually better results than BM4D, they appear to lose texture and smooth out the details of the results, that is, the building edge information is blurred. Thus, it is considerably difficult for these comparison methods to remove the heavy Poisson noise when the image is blurred. Figs. 1(i) and 2(i) show the performance of the proposed VSPT more detail in textured regions or edges and cleaner results in smooth regions are observed. The proposed VSPT better preserves the local details and removes the noise compared to the other methods.

2) Quantitative Comparison: To give a comprehensive quantitative evaluation of the denoising performance, the following three quantitative indices are employed to evaluate the results of the simulated experiments: peak SNR (PSNR), structure similarity (SSIM), and spectral angle map (SAM) [54]. The PSNR and SSIM are spatial-based indices, while the SAM is a spectral-based index. The values of PSNR and SSIM are directly proportional to the quality of the image, while the value of SAM is inversely proportional to the quality of the image. In other words, higher PSNR and SSIM values and a lower SAM value signify that the restored images are better.

Table I shows the overall quantitative evaluation results of the denoising methods for Pavia city center data. Table II shows the quantitative evaluation results of the different denoising methods for Washington DC Mall data. The best values for each quality index are indicated in bold. It can be shown from Tables I and II that the overall results are not very high, which is on account of the existence of blurs and the high Poisson noise, but the proposed VSPT achieves higher quantitative values than the other methods. This indicates that the proposed VSPT is superior in processing images that are blurred during imaging and contaminated by Poisson noise. This is consistent with the visual quality results shown above. In other words, using MAP estimation to build the denoising model is an effective alternative to the Anscombe transformation. In addition, we consider the presence of inevitable blurs in the imaging process, which can remove the Poisson noise more effectively.

3) Verified on HSIs of Large Size: With the development of hyperspectral remote sensing sensor technology and the improvement of academic exchange mechanism in China, hyperspectral remote sensing datasets have also been released in China, such as a hyperspectral dataset of Matiwan Village, Xiongan New District [56]. Acquired by the full-spectrum multimodal imaging spectrometer developed by the Shanghai Institute of Technical Physics for the high-resolution special aviation system, the spectral range of the aerial hyperspectral remote sensing image of Xiongan New Area (Matiwan Village) is 400–1000 nm, with 250 bands and a spatial resolution of 0.5 m. The image size is 3750 × 1580 pixels. To further verify the performance of the proposed method, we add a simulated experiment on Matiwan Village data, which is a group of HSI of large size. Subregion with the size of 1000 × 1000 × 250 is extracted for Matiwan Village data. We still normalize the gray value of each band of the HSI between [0, 1] before the simulation process. Since the test is carried out on a group of HSI of large size, we have

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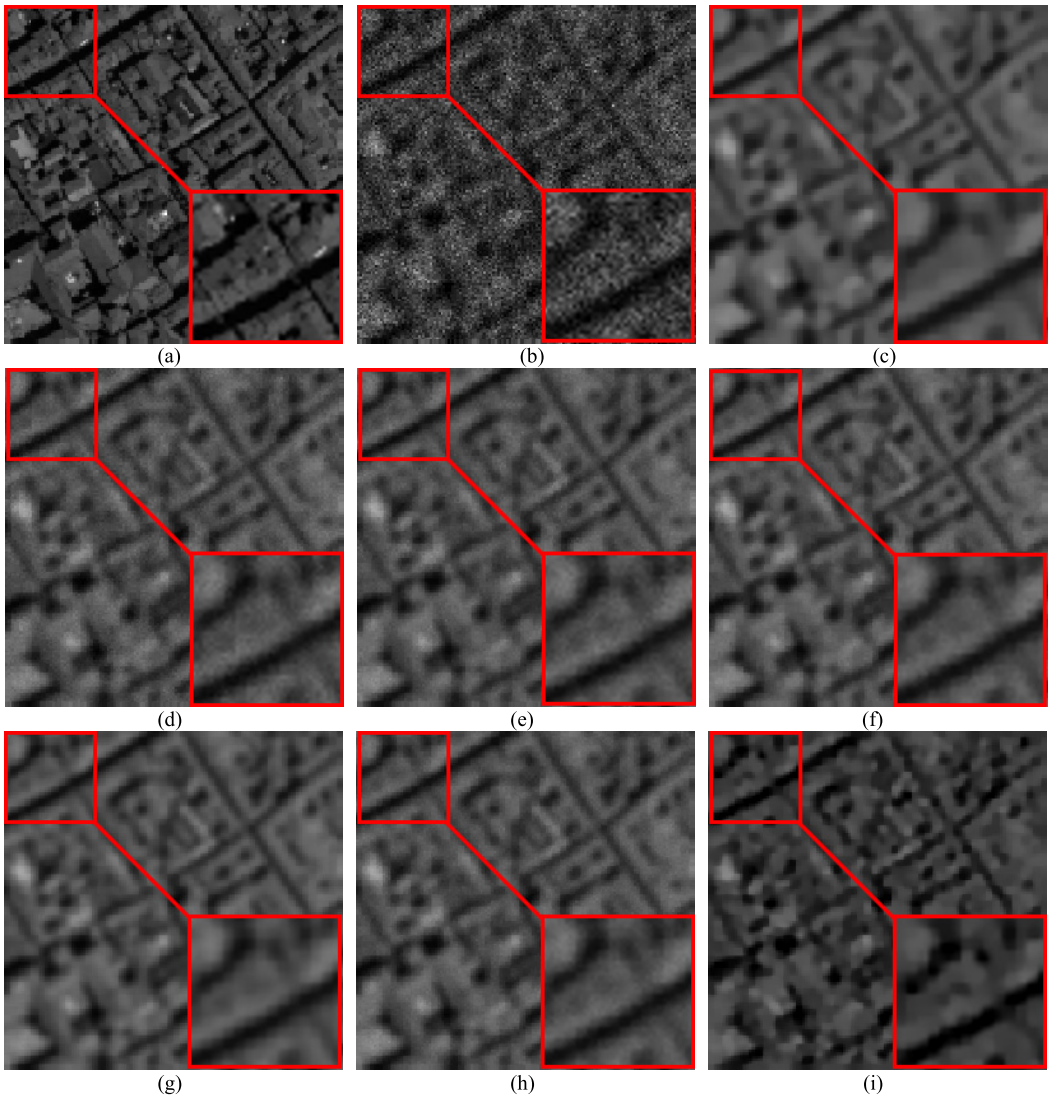


Fig. 1. Results on band 51 of Pavia city center data. (a) Band 51 as reference image, (b) Poissonian blurred image, (c) BM4D, (d) LRMR, (e) NAILRMA, (f) LRTDTV, (g) FastHyDe, (h) FGSLR, and (i) proposed VSPT.

TABLE I

QUANTITATIVE ASSESSMENT OF DIFFERENT DENOISING METHODS APPLIED TO PAVIA CITY CENTER DATA

	Noisy	BM4D	LRMR	NAILRMA	LRTDTV	FastHyDe	FGSLR	VSPT
PSNR	20.3326	20.1856	20.4132	20.4070	20.3700	20.5589	20.5407	**24.0678**
SSIM	0.2868	0.4996	0.5214	0.5289	0.5368	0.5523	0.5405	**0.5753**
SAM	0.4015	0.2878	0.2758	0.2750	0.2733	0.2693	0.2712	**0.2604**

TABLE II

QUANTITATIVE ASSESSMENT OF DIFFERENT DENOISING METHODS APPLIED TO WASHINGTON DC MALL DATA

	Noisy	BM4D	LRMR	NAILRMA	LRTDTV	FastHyDe	FGSLR	VSPT
PSNR	23.4134	23.9285	24.4486	24.3224	24.1043	24.4592	24.4806	**27.7176**
SSIM	0.3330	0.5238	0.5518	0.5522	0.5478	0.5615	0.5577	**0.6093**
SAM	0.5490	0.3933	0.3784	0.3777	0.3789	0.3738	0.3746	**0.3606**

correspondingly increased the noise and blur. We consider a Gaussian blur kernel with size 17×17 and standard deviation equal to 7 to simulate the possible blur generated during the imaging process. Then, we keep $a = 3.3 \times 10^{10}$ to add the Poisson distributed noise.

We also select one band in Matiwan Village data to present the visual effects of the different denoising methods. We take the value of λ as 19. Fig. 3 presents the visual effects of the different denoising methods for the simulated dataset. The original HSI of band 110 of the Matiwan Village

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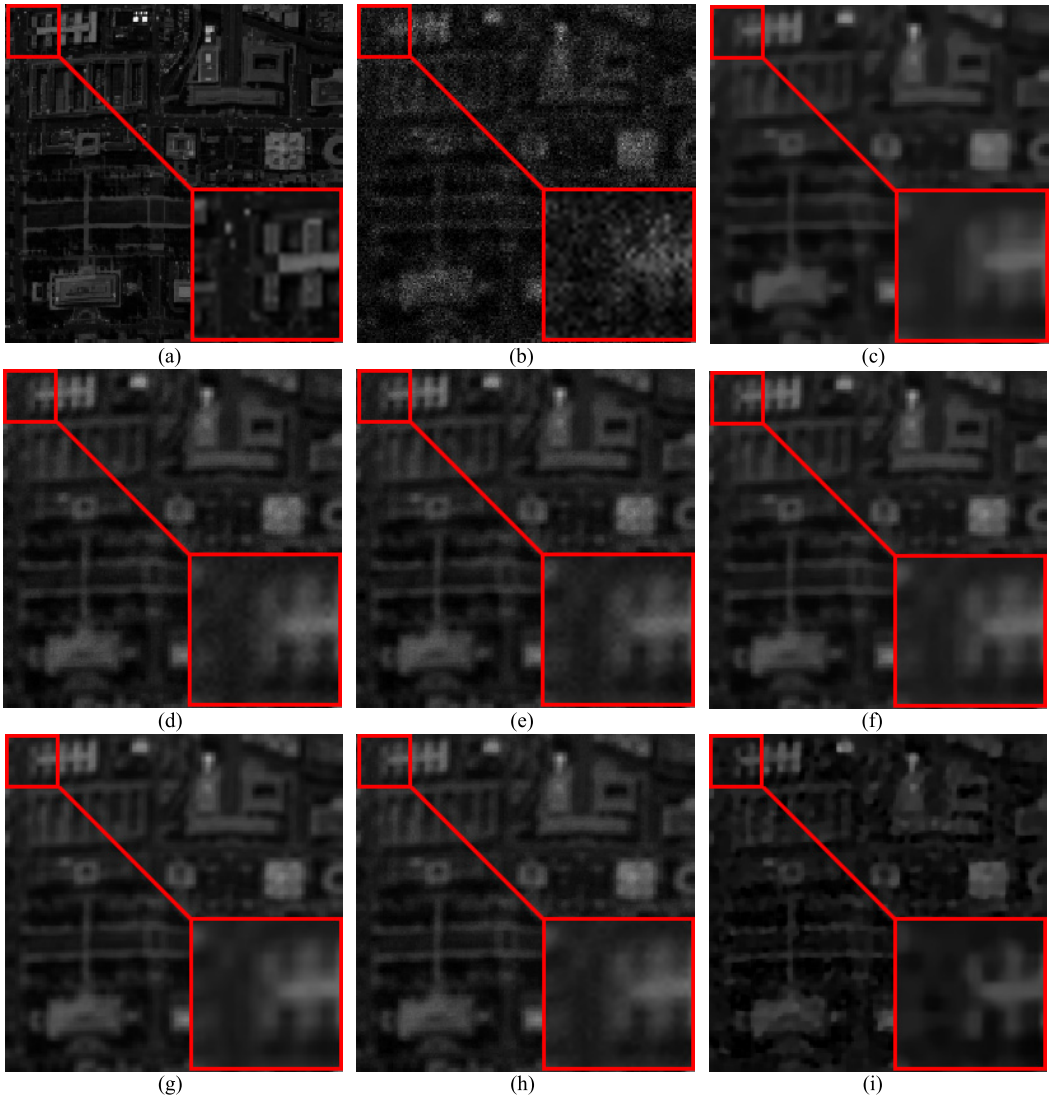


Fig. 2. Results on band 11 of Washington DC Mall data. (a) Band 11 as reference image, (b) Poissonian blurred image, (c) BM4D, (d) LRMR, (e) NAILRMA, (f) LRTDTV, (g) FastHyDe, (h) FGSLR, and (i) proposed VSPT.

TABLE III
QUANTITATIVE ASSESSMENT OF DIFFERENT DENOISING METHODS APPLIED TO XIONGAN DATA

	Noisy	BM4D	LRMR	NAILRMA	LRTDTV	FastHyDe	FGSLR	VSPT
PSNR	20.6055	20.8764	21.4051	21.3062	20.6158	21.5030	21.3186	**29.3897**
SSIM	0.6833	0.7975	0.7933	0.8006	0.7943	0.8147	0.8093	**0.8269**
SAM	0.3600	0.1413	0.1498	0.1421	0.1451	0.1369	0.1456	**0.1311**

data and the simulated degraded image are presented in Fig. 3(a) and (b), respectively. Fig. 3(c)–(i) displays the results of different denoising methods. Table II shows the quantitative evaluation results of the different denoising methods for Matiwan Village data. From Fig. 3(c)–(i), because of the simulation of a higher level of Poisson noise and the blurring effects of the imaging process, the visual information in the degraded bands is not very noticeable. However, we still find that the proposed method outperforms all the other algorithms and obtains the visually best result. It can be seen from the enlarged area in Fig. 3 that the edge of the image restored by our method is clearer in the case of large noise and blur. It can also be shown from Table III that the proposed VSPT achieves better quantitative values than the other methods. This indicates that the proposed VSPT is superior in processing images that are blurred during imaging and contaminated by Poisson noise. Visual quality and quantitative indexes are consistent. To sum up, our method still has advantages even when applied to a group of HSI of large size.

B. Real Data Experiments

The definition of noise type in the real world is sometimes actually subjective. In fact, we cannot really determine what kind of noise is polluting the real world image, that is to

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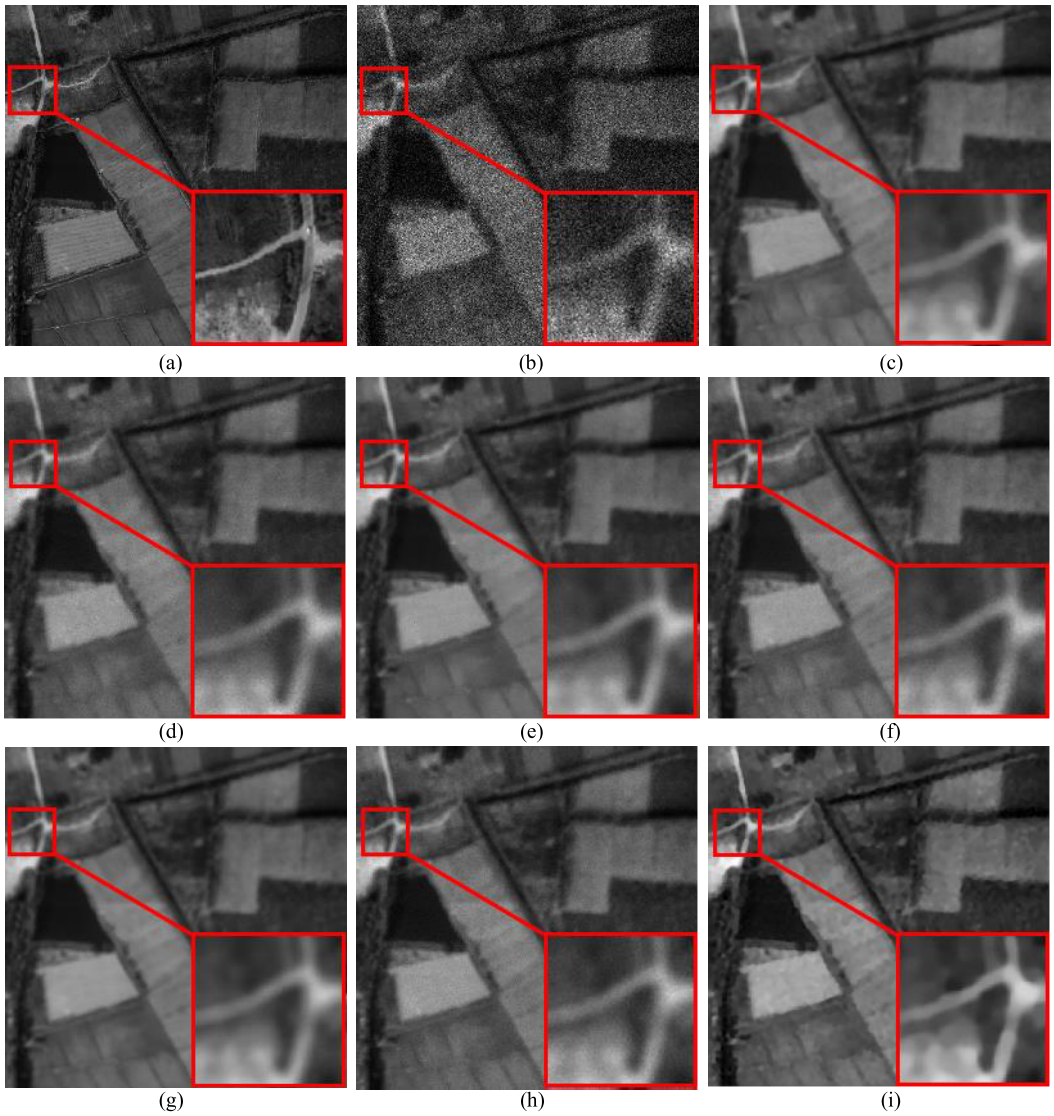


Fig. 3. Results on band 110 of Matiwan Village data. (a) Band 110 as reference image, (b) Poissonian blurred image, (c) BM4D, (d) LRMR, (e) NAILRMA, (f) LRTDTV, (g) FastHyDe, (h) FGSLR, and (i) proposed VSPT.

say, there is probably no prior knowledge about the nature of degradation in the reality situation. In order to further prove the applicability of the proposed VSPT in real life, we will evaluate its performance using real data, which has no prior knowledge regarding the nature of degradation. To ensure that the performance of each denoising method can be tested fairly in the experiment, we did not use the Anscombe transform, which may affect the final result in advance when using other contrast methods. We directly use other contrast methods to investigate their performance of real-world HSI denoising. Since there are no reference clean images, we just present the real Urban and Indian Pines images before and after denoising in Figs. 4 and 5.

HYDICE Urban data are used as real data [25] with a size equal to $307 \times 307 \times 210$. The data are degraded by various noise types and form a representative real dataset for testing data denoising methods. Although these data are mainly used to evaluate the denoising methods, we find that the images in some bands are blurred.

Fig. 4(a) shows a color image by combining three bands (205, 77, and 5). Since the blur generated in real-life imaging may not be large, we estimate a light blur kernel with a size of 2×2 and a standard deviation of 1. We take the value of λ as 1 in band 205, the value of λ as 100 in band 77, and the value of λ as 100 in band 5. Fig. 4(b) shows that the output of BM4D is worst and overly smoothed, and most of the information such as edges and textures is also missing. It is because it removes the noise by blurring the details to some extent. As shown in Fig. 4(c)–(g), the other comparison methods all perform better for HYDICE Urban data than BM4D and manage to recover most of the visual information, but it can be shown in red enlarged boxes that the visual effect obtained by the proposed VSPT in Fig. 4(h) is more pleasant. Because our method considers the inevitable blurring during the imaging process. So, in addition to removing noise, the results obtained by the proposed VSPT can make the structure of the edges richer and clearer by reducing the degree of blurring.

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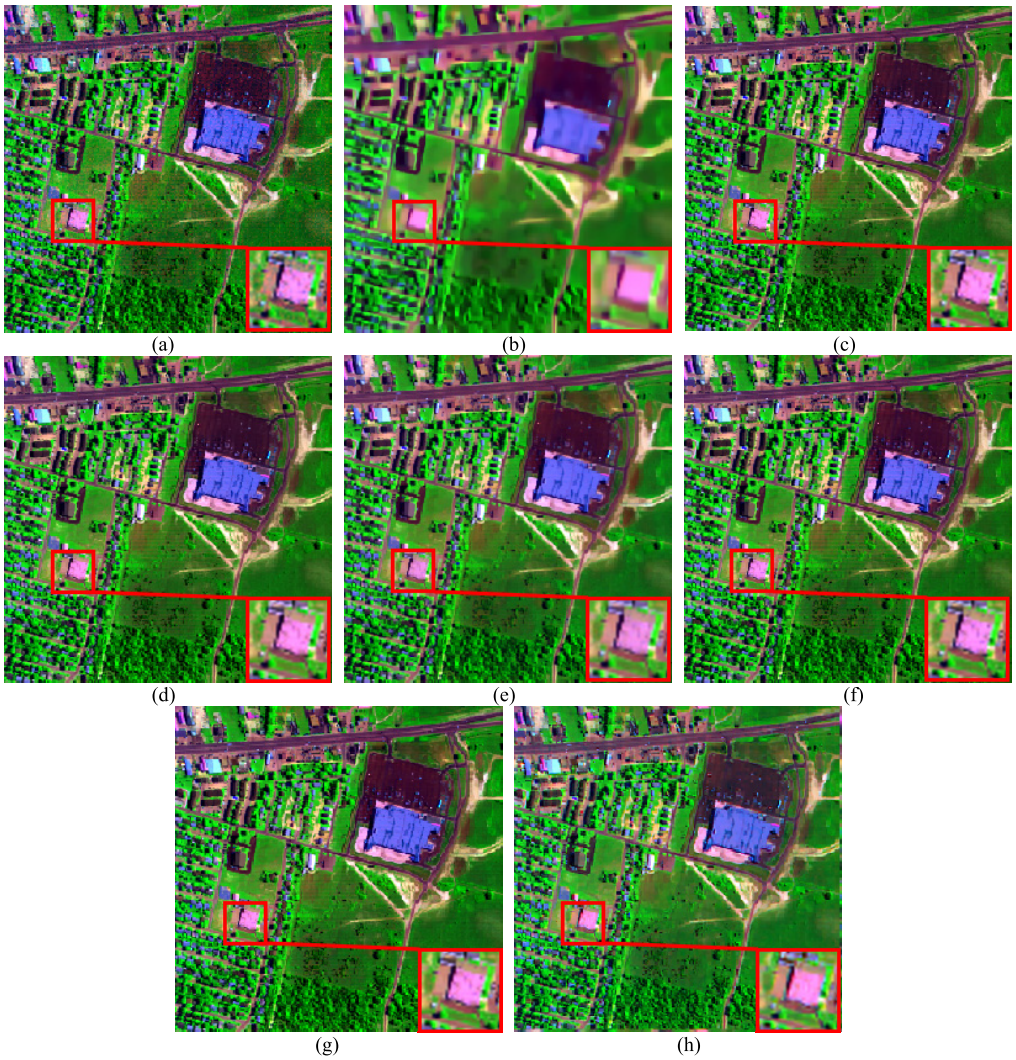


Fig. 4. Results on RGB bands (205, 77, and 5) of HYDICE Urban data. (a) RGB bands (205, 77, and 5) as reference image, (b) BM4D, (c) LRMR, (d) NAILRMA, (e) LRTDTV, (f) FastHyDe, (g) FGSLR, and (h) proposed VSPT.

The other one for real data is AVIRIS Indian Pines Data [21]. The size of data is 200 × 200 × 80, The image displays strong noise in a number of bands. We also find that the images in some bands are blurred. Fig. 5(a) shows a color image by combining three bands (102, 217, and 4). We still estimate a light blur kernel with a size of 145 × 145 × 220 and a standard deviation of 1. We take the value of λ as 150 in band 102, the value of λ as 1 in band 217, and the value of λ as 1 in band 4. As shown in red enlarged boxes in Fig. 5(b)–(h), our method can restore the edge more clearly than other methods on the basis of noise removal. At the same time, our color is closer to the original image, which to some extent shows that we also have protected spectral information [57]. We conclude from these results that although the proposed VSPT is designed in the Poisson noise and blurred assumption, it can also achieve the best results for real datasets. So, the proposed VSPT can be applicable in real life.

C. Discussion

1) Performance at Different Noise Levels: In this section, the performance of the proposed VSPT is investigated under different noise levels. To make a fair comparison, we consider a Gaussian blur kernel with size 8 × 8 and standard deviation equal to 3 to simulate the possible blur generated during the imaging process before testing different noise levels. In addition to $a = 1.0 \times 10^{10}$ (named as Case 1) in the previous simulation experiments, we select two other groups of a values, namely, $a = 2.4 \times 10^{10}$ (named as Case 2) and $a = 3.3 \times 10^{10}$ (named as Case 3) to test the performance of different denoising methods at different noise levels. For the Pavia city center data, the value of λ in Case 2 is 37 and the value of λ in Case 3 is 30. For the Washington DC Mall data, the value of λ in Case 2 is 23 and the value of λ in Case 3 is 15.

Fig. 6(a) and (b) shows the values of SSIM and SAM for Pavia city center data with different denoising methods. Fig. 7(a) and (b) shows the values of SSIM and SAM for Washington DC Mall data with different denoising methods. Summarizing the quantitative indices of these two data, we can conclude that the proposed VSPT is relatively robust for HSI denoising, as it can obtain the best quantitative results on different data under various noise conditions.

2) Performance at Different Blur Levels: Similarly, for a fair comparison, we keep $a = 1.0 \times 10^{10}$ before testing

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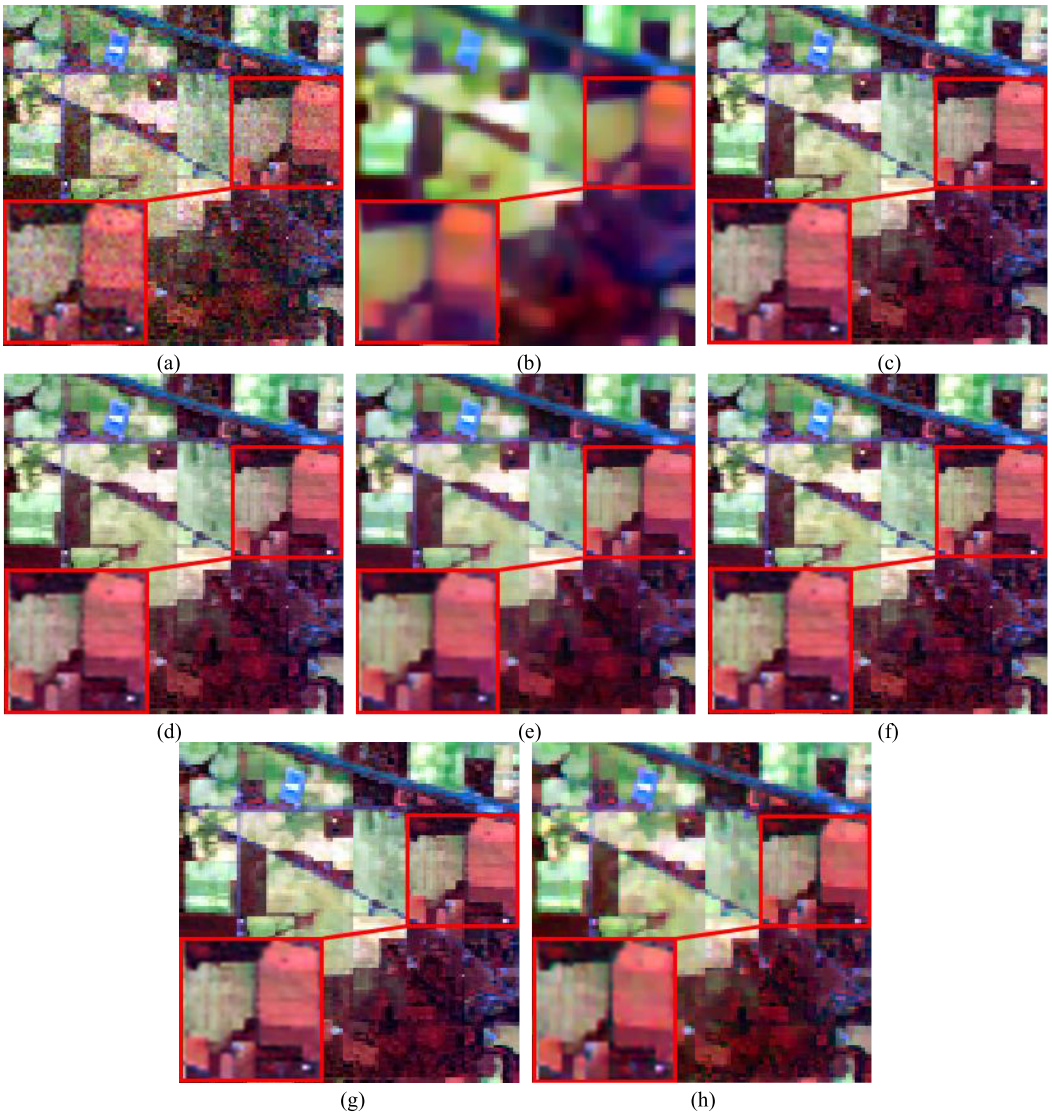


Fig. 5. Results on RGB bands (102, 217, and 4) of Indian Pines data. (a) RGB bands (102, 217, and 4) as reference image, (b) BM4D, (c) LRMR, (d) NAILRMA, (e) LRTDTV, (f) FastHyDe, (g) FGSLR, and (h) proposed VSPT.

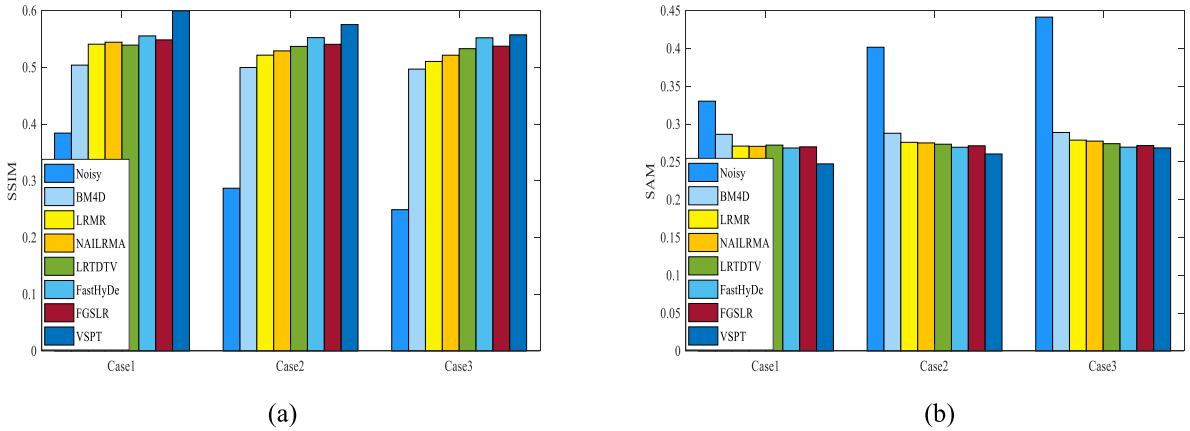
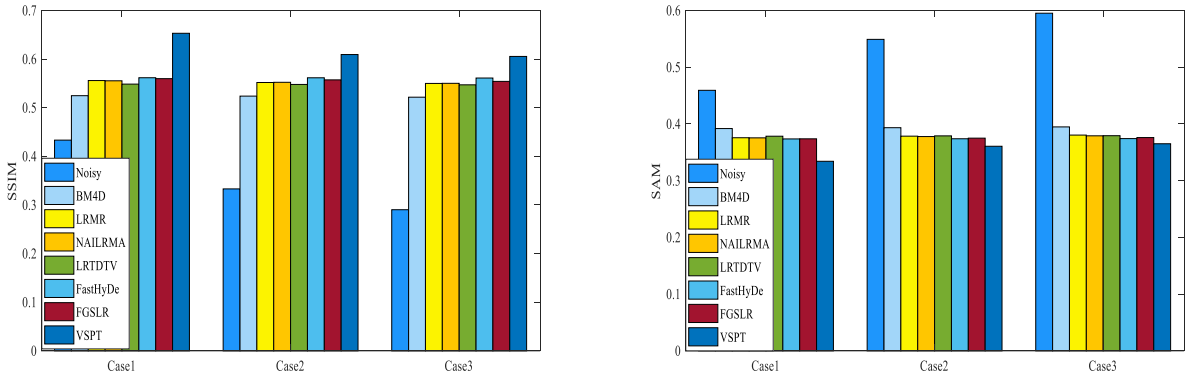


Fig. 6. Performance of different denoising methods for Pavia city center data at different noise levels. (a) SSIM value. (b) PSNR value.

different blur levels. We keep increasing the blur based on the simulation experiments. Three different cases are considered: Case 1 where the size of the blur kernel is 8 × 8 and the standard deviation is 3, Case 2 where the size of the blur kernel is 12 × 12 and the standard deviation is 5, and Case 3 where the size of the blur kernel is 17 × 17 and the standard

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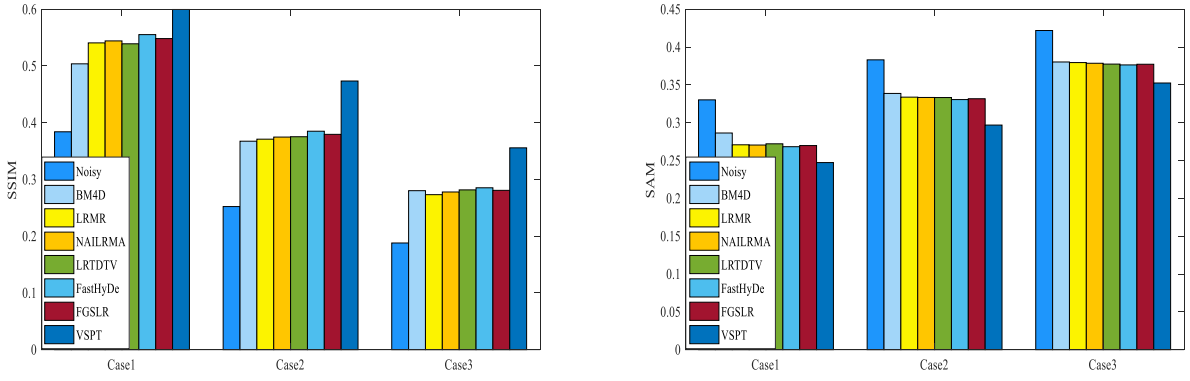


(a) (b)

Fig. 7. Performance of different denoising methods for Washington DC Mall data at different noise levels. (a) SSIM value. (b) PSNR value.

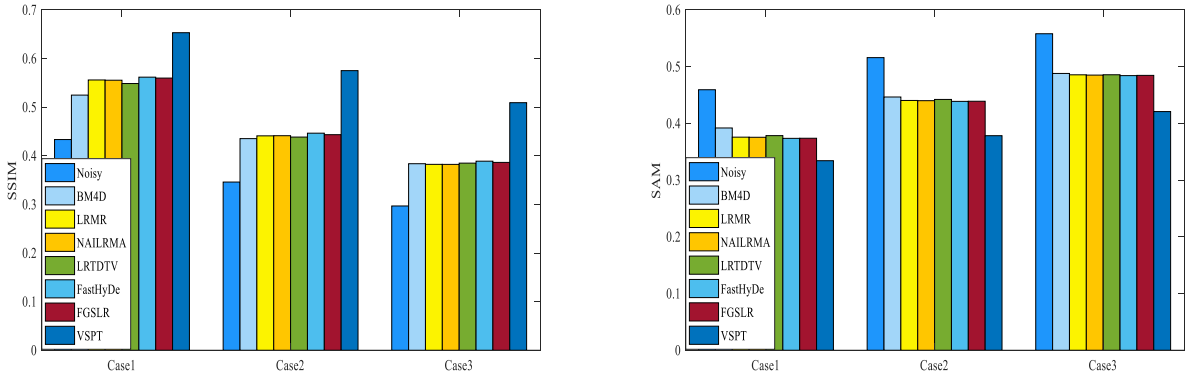
TABLE IV
COMPUTATIONAL TIME (S) OF DIFFERENT DENOISING METHODS

Data	BM4D	LRMR	NAILRMA	LRTDTV	FastHyDe	FGSLR	VSPT
Pavia city center	147.9011	44.1968	1.9930	97.7298	**0.7922**	74.8915	30.3048
Washington DC Mall	748.8125	178.4679	8.1155	471.9121	**2.0884**	584.6221	170.1333



(a) (b)

Fig. 8. Performance of different denoising methods for Pavia city center data at different blur levels. (a) SSIM value. (b) PSNR value.



(a) (b)

Fig. 9. Performance of different denoising methods for Washington DC Mall data at different blur levels. (a) SSIM value. (b) PSNR value.

deviation is 7. Therefore, we can simulate higher degrees of blur that may occur during the imaging process and test the performance of different denoising methods for different blur levels. For the Pavia city center data, the value of λ in Case 2 is 111 and the value of λ in Case 3 is 122. For the Washington DC Mall data, the value of λ in Case 2 is 69 and the value of λ in Case 3 is 93.

Fig. 8(a) and (b) shows the results for Pavia city center data with different denoising methods. Fig. 9(a) and (b) shows the values of SSIM and SAM for Washington DC

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Mall data with different denoising methods. It is obvious that the proposed VSPT achieves higher SSIM values and lower SAM values than other methods in almost every case, while the quantitative results of other comparison methods are not satisfactory. These results are sufficient to prove that the VSPT is more robust to various blur levels than other comparison methods. In other words, compared with other methods, the VSPT can simultaneously remove Poisson noise and deal with blur because it is based on the premise that the blur exists during the imaging process.

3) Computation Time: The computation time is another important performance indicator. Under the condition that $a = 1.0 \times 10^{10}$ and Gaussian blur kernel has a size of 8×8 and a standard deviation of 3, Table IV presents the computation times of different methods for Pavia city center data and Washington DC Mall data. The computation time of the proposed VSPT is only slightly higher than those of the simple FastHyDe and NAILRMA. This indicates that the VSPT can obtain the best results within an acceptable computation time.

IV. Conclusion

In this article, we proposed a Poissonian new blurred HSI denoising technique named as the VSPT in order to mitigate the problem caused by the simultaneous existence of Poisson noise and blur in real world. The technique could recover the blurred HSI contaminated by Poisson noise without using the Anscombe transform in advance. We characterized this problem under the MAP model by minimizing the combination of the negative logarithmic Poisson log-likelihood and the TV and used variable splitting and penalty method to solve this challenging optimization task. The final results were obtained by iteratively solving two subproblems involving TV regularization deconvolution and convex optimization. The proposed VSPT was tested using both simulated and real data experiments. The proposed method is shown to outperform the conventional denoising methods in terms of visual effects and quantitative evaluation.

Our proposed method can be improved further from the following four aspects.

1) Although the band-by-band processing can be refined further according to different situations, it will reduce some efficiency. Therefore, we will consider a weighting strategy of assigning different weights for each band according to the noise intensity, which can improve the overall processing efficiency.
2) In future work, we will also implement adaptive determination of the regularization parameters.
3) The TV regularization can be combined with low-rank regularizations to achieve some better results. We can look for some advanced low-rank regularizations to further improve the Poisson noise removal performance in the next step of work.
4) Deep learning-based methods have achieved SOTA performance. So, we will also consider to remove Poisson noise and blur in HSI by proposing the novel deep learning network in the future.

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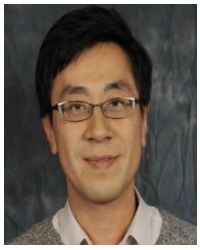
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